

Our residuals are this:

$$S = \sum_{p \neq q} \sum_{mn} d_{pq}^{mn} - g_p^{ma} M^{ab} \overline{g_q^{bn}}. \quad (1)$$

p and q are baselines, m and n are correlations ($\in r, l$). Repeated indices are summed. What we minimize is the norm of S , that is, $S\overline{S}$, or $|S|^2$. To this end, we compute gradient of the loss, that is, $\frac{\partial |S|^2}{\partial g_p^{mn}}$ for every m, n .

$$\begin{aligned} \frac{\partial |S|^2}{\partial g_p^{rr}} = & \sum_q \left(d_{pq}^{rr} \left(-g_q^{rl} \overline{m^{rl}} - g_q^{rr} \overline{m^{rr}} \right) + \right. \\ & d_{pq}^{rl} \left(-g_q^{ll} \overline{m^{rl}} - g_q^{lr} \overline{m^{rr}} \right) + \\ & g_p^{rr} \left(g_q^{ll} m^{rl} \overline{g_q^{ll} m^{rl}} + g_q^{ll} m^{rr} \overline{g_q^{lr} m^{rl}} + g_q^{lr} m^{rl} \overline{g_q^{ll} m^{rr}} + g_q^{lr} m^{rr} \overline{g_q^{lr} m^{rr}} + g_q^{rl} m^{rl} \overline{g_q^{rl} m^{rl}} + g_q^{rl} m^{rr} \overline{g_q^{rr} m^{rl}} + g_q^{rr} m^{rl} \overline{g_q^{rl} m^{rr}} + g_q^{rr} m^{rr} \overline{g_q^{rr} m^{rr}} \right) + \\ & \left. g_p^{rl} \left(g_q^{ll} m^{ll} \overline{g_q^{ll} m^{rl}} + g_q^{ll} m^{lr} \overline{g_q^{lr} m^{rl}} + g_q^{lr} m^{ll} \overline{g_q^{ll} m^{rr}} + g_q^{lr} m^{lr} \overline{g_q^{lr} m^{rr}} + g_q^{rl} m^{ll} \overline{g_q^{rl} m^{rl}} + g_q^{rl} m^{lr} \overline{g_q^{rr} m^{rl}} + g_q^{rr} m^{ll} \overline{g_q^{rl} m^{rr}} + g_q^{rr} m^{lr} \overline{g_q^{rr} m^{rr}} \right) \right). \end{aligned} \quad (2)$$

$$\begin{aligned} \frac{\partial |S|^2}{\partial g_p^{rl}} = & \sum_q \left(d_{pq}^{rr} \left(-g_q^{rl} \overline{m^{ll}} - g_q^{rr} \overline{m^{lr}} \right) + \right. \\ & d_{pq}^{rl} \left(-g_q^{ll} \overline{m^{ll}} - g_q^{lr} \overline{m^{lr}} \right) + \\ & g_p^{rr} \left(g_q^{ll} m^{rl} \overline{g_q^{ll} m^{ll}} + g_q^{ll} m^{rr} \overline{g_q^{lr} m^{ll}} + g_q^{lr} m^{rl} \overline{g_q^{ll} m^{lr}} + g_q^{lr} m^{rr} \overline{g_q^{lr} m^{lr}} + g_q^{rl} m^{rl} \overline{g_q^{rl} m^{ll}} + g_q^{rl} m^{rr} \overline{g_q^{rr} m^{ll}} + g_q^{rr} m^{rl} \overline{g_q^{rl} m^{lr}} + g_q^{rr} m^{rr} \overline{g_q^{rr} m^{lr}} \right) + \\ & \left. g_p^{rl} \left(g_q^{ll} m^{ll} \overline{g_q^{ll} m^{ll}} + g_q^{ll} m^{lr} \overline{g_q^{lr} m^{ll}} + g_q^{lr} m^{ll} \overline{g_q^{ll} m^{lr}} + g_q^{lr} m^{lr} \overline{g_q^{lr} m^{lr}} + g_q^{rl} m^{ll} \overline{g_q^{rl} m^{ll}} + g_q^{rl} m^{lr} \overline{g_q^{rr} m^{ll}} + g_q^{rr} m^{ll} \overline{g_q^{rl} m^{lr}} + g_q^{rr} m^{lr} \overline{g_q^{rr} m^{lr}} \right) \right). \end{aligned} \quad (3)$$

$$\begin{aligned} \frac{\partial |S|^2}{\partial g_p^{lr}} = & \sum_q \left(d_{pq}^{lr} \left(-g_q^{rl} \overline{m^{rl}} - g_q^{rr} \overline{m^{rr}} \right) + \right. \\ & d_{pq}^{ll} \left(-g_q^{ll} \overline{m^{rl}} - g_q^{lr} \overline{m^{rr}} \right) + \\ & g_p^{lr} \left(g_q^{ll} m^{rl} \overline{g_q^{ll} m^{rl}} + g_q^{ll} m^{rr} \overline{g_q^{lr} m^{rl}} + g_q^{lr} m^{rl} \overline{g_q^{ll} m^{rr}} + g_q^{lr} m^{rr} \overline{g_q^{lr} m^{rr}} + g_q^{rl} m^{rl} \overline{g_q^{rl} m^{rl}} + g_q^{rl} m^{rr} \overline{g_q^{rr} m^{rl}} + g_q^{rr} m^{rl} \overline{g_q^{rl} m^{rr}} + g_q^{rr} m^{rr} \overline{g_q^{rr} m^{rr}} \right) + \\ & \left. g_p^{ll} \left(g_q^{ll} m^{ll} \overline{g_q^{ll} m^{rl}} + g_q^{ll} m^{lr} \overline{g_q^{lr} m^{rl}} + g_q^{lr} m^{ll} \overline{g_q^{ll} m^{rr}} + g_q^{lr} m^{lr} \overline{g_q^{lr} m^{rr}} + g_q^{rl} m^{ll} \overline{g_q^{rl} m^{rl}} + g_q^{rl} m^{lr} \overline{g_q^{rr} m^{rl}} + g_q^{rr} m^{ll} \overline{g_q^{rl} m^{rr}} + g_q^{rr} m^{lr} \overline{g_q^{rr} m^{rr}} \right) \right). \end{aligned} \quad (4)$$

$$\begin{aligned}
\frac{\partial |S|^2}{\partial g_p^{ll}} = & \sum_q \left(d_{pq}^{lr} \left(-g_q^{rl} \overline{m^{ll}} - g_q^{rr} \overline{m^{lr}} \right) + \right. \\
& d_{pq}^{ll} \left(-g_q^{ll} \overline{m^{ll}} - g_q^{lr} \overline{m^{lr}} \right) + \\
& g_p^{lr} \left(g_q^{ll} m^{rl} \overline{g_q^{ll} m^{ll}} + g_q^{ll} m^{rr} \overline{g_q^{lr} m^{ll}} + g_q^{lr} m^{rl} \overline{g_q^{ll} m^{lr}} + g_q^{lr} m^{rr} \overline{g_q^{lr} m^{lr}} + g_q^{rl} m^{rl} \overline{g_q^{rl} m^{ll}} + g_q^{rl} m^{rr} \overline{g_q^{rr} m^{ll}} + g_q^{rr} m^{rl} \overline{g_q^{rl} m^{lr}} + g_q^{rr} m^{rr} \overline{g_q^{rr} m^{lr}} \right) + \\
& \left. g_p^{ll} \left(g_q^{ll} m^{ll} \overline{g_q^{ll} m^{ll}} + g_q^{ll} m^{lr} \overline{g_q^{lr} m^{ll}} + g_q^{lr} m^{ll} \overline{g_q^{ll} m^{lr}} + g_q^{lr} m^{lr} \overline{g_q^{lr} m^{lr}} + g_q^{rl} m^{ll} \overline{g_q^{rl} m^{ll}} + g_q^{rl} m^{lr} \overline{g_q^{rr} m^{ll}} + g_q^{rr} m^{ll} \overline{g_q^{rl} m^{lr}} + g_q^{rr} m^{lr} \overline{g_q^{rr} m^{lr}} \right) \right). \tag{5}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial |S|^2}{\partial g_q^{rr}} = & \sum_p \left((-g_p^{ll} m^{lr} - g_p^{lr} m^{rr}) \overline{d_{pq}^{lr}} + \right. \\
& (-g_p^{rl} m^{lr} - g_p^{rr} m^{rr}) \overline{d_{pq}^{rr}} + \\
& g_q^{rl} \left(g_p^{ll} m^{lr} \overline{g_p^{ll} m^{ll}} + g_p^{ll} m^{lr} \overline{g_p^{lr} m^{rl}} + g_p^{lr} m^{rr} \overline{g_p^{ll} m^{ll}} + g_p^{lr} m^{rr} \overline{g_p^{lr} m^{rl}} + g_p^{rl} m^{lr} \overline{g_p^{rl} m^{ll}} + g_p^{rl} m^{lr} \overline{g_p^{rr} m^{rl}} + g_p^{rr} m^{rr} \overline{g_p^{rl} m^{ll}} + g_p^{rr} m^{rr} \overline{g_p^{rr} m^{rl}} \right) + \\
& \left. g_q^{rr} \left(g_p^{ll} m^{lr} \overline{g_p^{ll} m^{lr}} + g_p^{ll} m^{lr} \overline{g_p^{lr} m^{rr}} + g_p^{lr} m^{rr} \overline{g_p^{ll} m^{lr}} + g_p^{lr} m^{rr} \overline{g_p^{lr} m^{rr}} + g_p^{rl} m^{lr} \overline{g_p^{rl} m^{lr}} + g_p^{rl} m^{lr} \overline{g_p^{rr} m^{rr}} + g_p^{rr} m^{rr} \overline{g_p^{rl} m^{lr}} + g_p^{rr} m^{rr} \overline{g_p^{rr} m^{rr}} \right) \right). \tag{6}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial |S|^2}{\partial g_q^{rl}} = & \sum_p \left((-g_p^{ll} m^{ll} - g_p^{lr} m^{rl}) \overline{d_{pq}^{lr}} + \right. \\
& (-g_p^{rl} m^{ll} - g_p^{rr} m^{rl}) \overline{d_{pq}^{rr}} + \\
& g_q^{rl} \left(g_p^{ll} m^{ll} \overline{g_p^{ll} m^{ll}} + g_p^{ll} m^{ll} \overline{g_p^{lr} m^{rl}} + g_p^{lr} m^{rl} \overline{g_p^{ll} m^{ll}} + g_p^{lr} m^{rl} \overline{g_p^{lr} m^{rl}} + g_p^{rl} m^{ll} \overline{g_p^{rl} m^{ll}} + g_p^{rl} m^{ll} \overline{g_p^{rr} m^{rl}} + g_p^{rr} m^{rl} \overline{g_p^{rl} m^{ll}} + g_p^{rr} m^{rl} \overline{g_p^{rr} m^{rl}} \right) + \\
& \left. g_q^{rr} \left(g_p^{ll} m^{ll} \overline{g_p^{ll} m^{lr}} + g_p^{ll} m^{ll} \overline{g_p^{lr} m^{rr}} + g_p^{lr} m^{rl} \overline{g_p^{ll} m^{lr}} + g_p^{lr} m^{rl} \overline{g_p^{lr} m^{rr}} + g_p^{rl} m^{ll} \overline{g_p^{rl} m^{lr}} + g_p^{rl} m^{ll} \overline{g_p^{rr} m^{rr}} + g_p^{rr} m^{rl} \overline{g_p^{rl} m^{lr}} + g_p^{rr} m^{rl} \overline{g_p^{rr} m^{rr}} \right) \right). \tag{7}
\end{aligned}$$

$$\begin{aligned}
\frac{\partial |S|^2}{\partial g_q^{lr}} &= \sum_p \\
&(-g_p^{ll} m^{lr} - g_p^{lr} m^{rr}) \overline{d_{pq}^{ll}} + \\
&(-g_p^{rl} m^{lr} - g_p^{rr} m^{rr}) \overline{d_{pq}^{rl}} + \\
&g_q^{ll} \left(g_p^{ll} m^{lr} \overline{g_p^{ll} m^{ll}} + g_p^{ll} m^{lr} \overline{g_p^{lr} m^{rl}} + g_p^{lr} m^{rr} \overline{g_p^{ll} m^{ll}} + g_p^{lr} m^{rr} \overline{g_p^{lr} m^{rl}} + g_p^{rl} m^{lr} \overline{g_p^{rl} m^{ll}} + g_p^{rl} m^{lr} \overline{g_p^{rr} m^{rl}} + g_p^{rr} m^{rr} \overline{g_p^{rl} m^{ll}} + g_p^{rr} m^{rr} \overline{g_p^{rr} m^{rl}} \right) + \\
&g_q^{lr} \left(g_p^{ll} m^{lr} \overline{g_p^{ll} m^{lr}} + g_p^{ll} m^{lr} \overline{g_p^{lr} m^{rr}} + g_p^{lr} m^{rr} \overline{g_p^{ll} m^{lr}} + g_p^{lr} m^{rr} \overline{g_p^{lr} m^{rr}} + g_p^{rl} m^{lr} \overline{g_p^{rl} m^{lr}} + g_p^{rl} m^{lr} \overline{g_p^{rr} m^{rr}} + g_p^{rr} m^{rr} \overline{g_p^{rl} m^{lr}} + g_p^{rr} m^{rr} \overline{g_p^{rr} m^{rr}} \right).
\end{aligned} \tag{8}$$

$$\begin{aligned}
\frac{\partial |S|^2}{\partial g_q^{ll}} &= \sum_p \\
&(-g_p^{ll} m^{ll} - g_p^{lr} m^{rl}) \overline{d_{pq}^{ll}} + \\
&(-g_p^{rl} m^{ll} - g_p^{rr} m^{rl}) \overline{d_{pq}^{rl}} + \\
&g_q^{ll} \left(g_p^{ll} m^{ll} \overline{g_p^{ll} m^{ll}} + g_p^{ll} m^{ll} \overline{g_p^{lr} m^{rl}} + g_p^{lr} m^{rl} \overline{g_p^{ll} m^{ll}} + g_p^{lr} m^{rl} \overline{g_p^{lr} m^{rl}} + g_p^{rl} m^{ll} \overline{g_p^{rl} m^{ll}} + g_p^{rl} m^{ll} \overline{g_p^{rr} m^{rl}} + g_p^{rr} m^{rl} \overline{g_p^{rl} m^{ll}} + g_p^{rr} m^{rl} \overline{g_p^{rr} m^{rl}} \right) + \\
&g_q^{lr} \left(g_p^{ll} m^{ll} \overline{g_p^{ll} m^{lr}} + g_p^{ll} m^{ll} \overline{g_p^{lr} m^{rr}} + g_p^{lr} m^{rl} \overline{g_p^{ll} m^{lr}} + g_p^{lr} m^{rl} \overline{g_p^{lr} m^{rr}} + g_p^{rl} m^{ll} \overline{g_p^{rl} m^{lr}} + g_p^{rl} m^{ll} \overline{g_p^{rr} m^{rr}} + g_p^{rr} m^{rl} \overline{g_p^{rl} m^{lr}} + g_p^{rr} m^{rl} \overline{g_p^{rr} m^{rr}} \right).
\end{aligned} \tag{9}$$